

Swinburne University of Technology

School of Science, Computing and Emerging Technologies

Contribution Document

Subject Code: SWE30003

Unit Title: Software Architectures and Design

Assignment number: 03

Assignment Title: Design Implementation

Due date: Aug 5th, 2026

Tutorial Day and Time: Mondays, 13:00 - 16:00 p.m

Project Group: SG3_3

Tutor: Mr.Pham Thai Ky Trung

1. **Time spent (hours):** fill in the time spent on each significant work item in the following table.

Work Item	Truong Ngoc Gia Hieu	Lu Nhat Hoang	Duong Nguyen Dang
Project Scope, Domain Vocabulary and Analysis	3	4	5
Architecture, CQRS and DB Design (ERD)	6	2	3
Tasks and Support Specification (Parts D and Functional Requirements)	5	6	6
CRUD Validation Matrix and Quality Attributes	2	3	5
Workflows and Activity Diagrams (Part E)	2	6	3
Final Integration, Formatting and QA	5	2	2
Total	23	23	24

Description of contributions:

- **Truong Ngoc Gia Hieu:** Led system architectural design (CQRS, Read Replicas, Pessimistic Locking) and created the Entity Relationship Diagram (ERD). Authored task specifications for *Fleet Asset Management* (Task 4.3) and *Real-Time Transit Tracking* (Task 4.6). Responsible for report integration, document formatting, and overall quality assurance.
- **Lu Nhat Hoang:** Mapped regional hub operations and financial settlement workflows. Authored task specifications for *Branch & Hub Setup* (Task 4.2), *Process Order Modifications* (Task 4.8), *Automated Invoice Generation* (Task 4.9), and *Payment Reconciliation* (Task 4.10). Designed operational Activity Diagrams and workflow models.
- **Duong Nguyen Dang:** Conducted domain vocabulary and enterprise pain point analysis. Authored task specifications for *User & Access Management* (Task 4.1), *Order Management* (Task 4.4), *Shipment Scheduling & Dispatch* (Task 4.5), and *Handle Delivery Exceptions* (Task 4.7). Co-developed the CRUD validation matrix and quality attribute scenarios.

2. Evidence of collaboration / teamwork on each significant work item :

SWE30003

Tệp Chỉnh sửa Xem Chèn Định dạng Công cụ Tiện ích Trợ giúp Trợ năng

Trình đơn 100% Văn bản Times 12 B I U A

Chia sẻ Nâng cấp

Các thẻ trong tài liệu

- Group Formation
- Weekly Tutorial
 - Week 2
 - Week 3
 - Week 4
 - Week 5
 - Week 6
 - Week 7
 - Week 8
 - Week 9
- Assignments
 - Assignment1
 - Requirements
 - Analyzing
 - Scenario 2
 - Report 1
 - Assignment2
 - Report02

keep a copy for yourself for future reference. (It is required that all students in a group should be from the same tutorial)

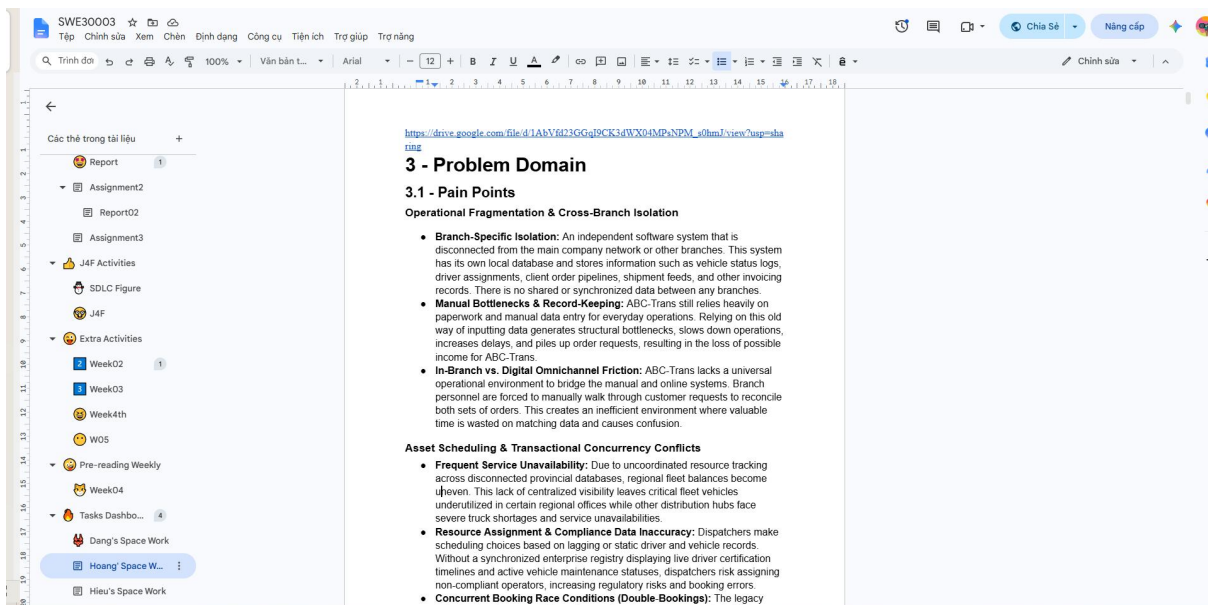
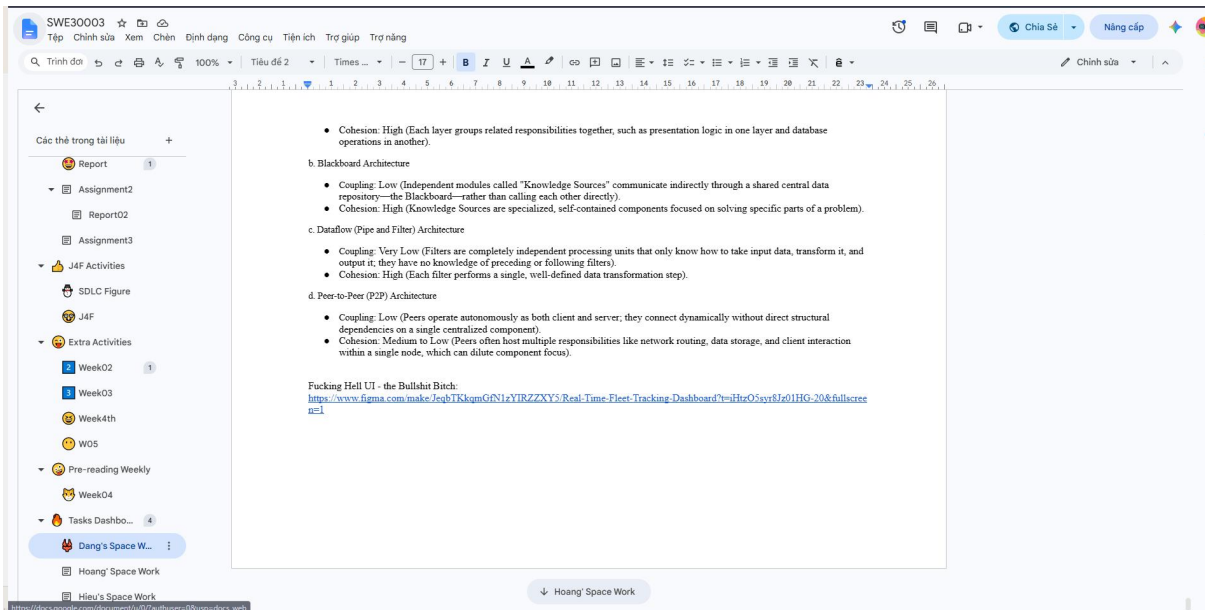
Tutorial session (day and time): Mondays, 13:00 - 16:00 p.m

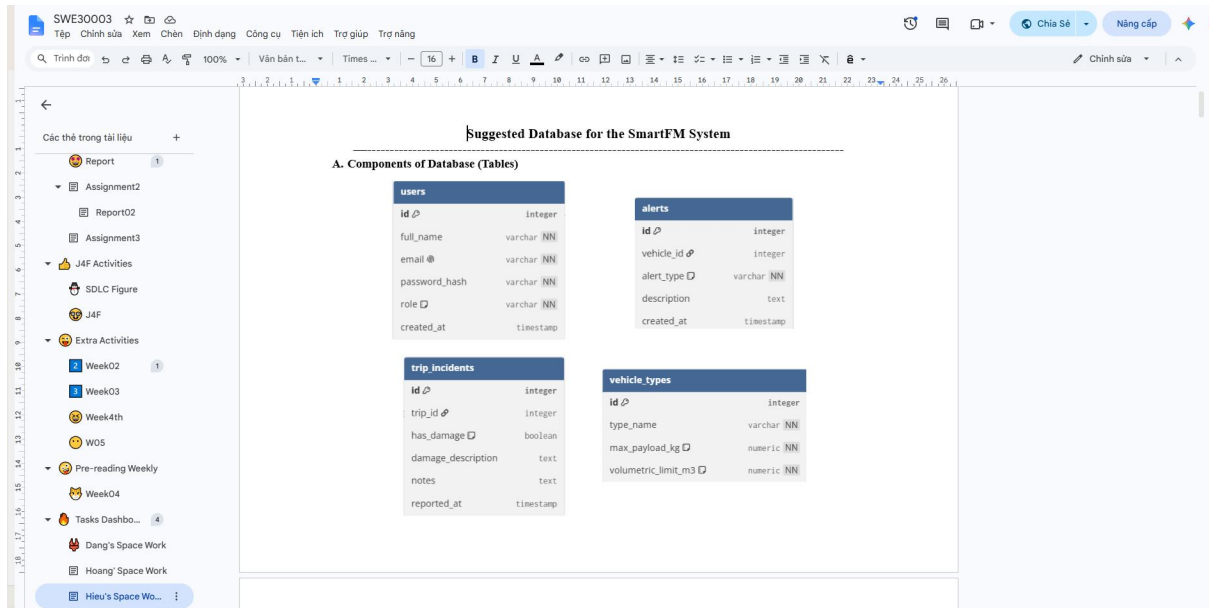
Student-ID	Name	Signature
105565520	Truong Ngoc Gia Hieu	
105508444	Duong Nguyen Dang	
105234956	Lu Nhat Hoang	

Primary group contact person:

Name: Truong Ngoc Gia Hieu
Email: 105565520@student.swin.edu.au

Weekly Tutorial





3. Signed by all members:

Student-ID

Name

Signature

105565520

Truong Ngoc Gia Hieu

105508444

Duong Nguyen Dang

105234956

Lu Nhat Hoang